

PERIPHERAL BENZEL TILINGS WITH RIGHT STONES AND THREE BONE ORIENTATIONS

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Formalization status. A kernel-only Lean 4 formalization of the results in this manuscript has been completed using Mathlib, with no project-specific axioms.

ABSTRACT. Propp asked for the number of tilings of the peripheral benzel $\mathcal{B}(n, 2n - 3)$ by right stones and all three orientations of bones. We prove his conjectured formula

$$T_{103}(n, 2n - 3) = \frac{(3n + 3)(3n - 7)!}{(n - 5)!(2n - 1)!} \quad (n \geq 5).$$

The proof first partitions the hexagonal cells into periodic three-cell owners. A linear owner-label energy forces every stone to occupy one owner and every bone to become a label-preserving directed edge. The edge set is then a directed Y : three paths from the deficient corner owners to one interior sink, with no other common vertices. At a fixed sink, the two last-edge chiralities reduce to independent one-dimensional ballot counts; in one chirality, three adjacent-binomial factors cancel cyclically. A three-variable constant-term evaluation followed by Lagrange inversion gives the stated formula.

1. INTRODUCTION

Benzels are threefold-symmetric regions in the hexagonal grid introduced by Propp as finite regions for exact trimer enumeration [6]. The five relevant trihexes are two triangular *stones* and three collinear *bones*. Write $T_{ijk}(a, b)$ for the number of tilings of the (a, b) -benzel when right stones are allowed if $i = 1$, left stones are allowed if $j = 1$, and k bone orientations are allowed.

Several of Propp’s problems have been solved using abacus and compression bijections [3, 2]. Those methods are particularly effective when at least one bone orientation is forbidden. The present problem permits all three bone orientations and lies on the peripheral diagonal $b = 2a - 3$. Propp states the following as Problem 6 and reports verification for $5 \leq n \leq 16$ [6]. Propp’s online status table, whose most recent problem update is dated June 2025, still listed Problem 6 as open when accessed on 22 August 2026 [7].

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Theorem 1.1 (Propp’s Problem 6). *For every integer $n \geq 5$,*

$$T_{103}(n, 2n - 3) = \frac{(3n + 3)(3n - 7)!}{(n - 5)!(2n - 1)!}.$$

The proof uses a periodic right-stone tiling to partition the plane into three-cell owners. Although the benzel has quadratic area, the Conway–Lagarias invariant and an owner-label energy force every admissible tiling to differ from this periodic background along only three directed paths. These paths have total length $3n - 6$ and meet at a single interior owner. The rest of the proof enumerates the three-path configurations.

Put $m = n - 5$. The conjectured number has the useful ballot form

$$(1) \quad 2 \binom{3m + 8}{m} - \binom{3m + 8}{m - 1} = \frac{3(m + 6)}{2m + 9} \binom{3m + 8}{m}.$$

The path reduction explains both terms: one sink chirality gives a cyclically telescoping product of binomial counts, while the other gives ballot counts.

2. BENZELS, STONES, AND BONES

Represent a hexagonal cell center by $(i, j) \in \mathbb{Z}^2$ and put $k = 1 - i - j$. This is the barycentric convention used by Propp [6]. The (a, b) -benzel is the cell set

$$\mathcal{B}(a, b) = \{(i, j) : -(a - 1) \leq j - i, k - j, i - k \leq b - 1\}.$$

The right stone is the translate class of

$$R = \{(0, 0), (1, 0), (0, 1)\},$$

and the three bones have direction sets

$$H = \{(0, 0), (1, 0), (2, 0)\},$$

$$V = \{(0, 0), (0, 1), (0, 2)\},$$

$$D = \{(0, 0), (1, -1), (2, -2)\}.$$

Only translates, not rotations beyond these three listed bone types, are used.

The area and Conway–Lagarias invariant give the following tile counts [1, 6].

Lemma 2.1. *Every type-103 tiling of $\mathcal{B}(n, 2n - 3)$ has*

$$\frac{(n - 5)(n - 2)}{2} \text{ right stones and } 3(n - 2) \text{ bones.}$$

Proof. Substitution in the benzel area formula gives

$$|\mathcal{B}(n, 2n - 3)| = \frac{3(n - 2)(n + 1)}{2},$$

so a tiling has $(n - 2)(n + 1)/2$ trihexes. Since no left stones are allowed, the Conway–Lagarias invariant equals the number of right stones. Its $a + b \equiv 0$

(mod 3) formula at $(a, b) = (n, 2n - 3)$ is

$$\frac{3(a - b)^2 - a - b}{6} = \frac{(n - 5)(n - 2)}{2}.$$

Subtracting from the total number of trihexes gives $3(n - 2)$ bones. $\square$

3. PERIODIC OWNERS AND THE ENERGY CERTIFICATE

Put $t = n - 2$ and work modulo three. An *owner anchor* is a pair (q, r) with

$$q - r \equiv t \pmod{3}.$$

The anchor owns the three cells

$$(q, r), \quad (q + 1, r), \quad (q, r + 1),$$

labelled 0, 1, 2, respectively. Every cell of the infinite hexagonal grid has a unique owner: among its three possible right-stone anchors, the three values of $q - r$ represent the three residue classes.

For an owner anchor (q, r) define

$$(2) \quad u = \frac{t - q + r}{3}, \quad v = \frac{t - q - 2r}{3}, \quad w = \frac{t + 2q + r}{3}.$$

These are integers with $u + v + w = t$.

Lemma 3.1 (Owner simplex). *The owners meeting $\mathcal{B}(n, 2n - 3)$ are in bijection with*

$$\Delta_t = \{(u, v, w) \in \mathbb{N}_0^3 : u + v + w = t\}.$$

Every non-corner owner contributes all three of its cells to the benzel. The three corner owners contribute two cells each: $(0, t, 0)$ misses label 0, $(0, 0, t)$ misses label 1, and $(t, 0, 0)$ misses label 2.

Proof. For a cell (i, j) put $k = 1 - i - j$ as above. Substitution of the three labels at (q, r) gives the following ordered triples $(j - i, k - j, i - k)$:

label	$j - i$	$k - j$	$i - k$
0	$3u - t$	$3v - t + 1$	$3w - t - 1$
1	$3u - t - 1$	$3v - t$	$3w - t + 1$
2	$3u - t + 1$	$3v - t - 1$	$3w - t$

The benzel bounds are $-t - 1 \leq j - i, k - j, i - k \leq 2t$. If any one of the three cells lies in the benzel, its row in the table, together with the integrality of u, v, w , forces $u, v, w \geq 0$. Conversely, suppose $u, v, w \geq 0$ and $u + v + w = t$. The table then shows

$$\begin{aligned} \text{label 0 is present} &\iff v < t, \\ \text{label 1 is present} &\iff w < t, \\ \text{label 2 is present} &\iff u < t. \end{aligned}$$

The inverse coordinate change is

$$q = w - u, \quad r = u - v;$$

indeed $q - r = t - 3u \equiv t \pmod{3}$. Thus every owner meeting the benzel corresponds to a unique point of Δ_t , and every point of Δ_t contributes at least two cells. A coordinate can equal t only when the other two coordinates vanish. The three such points give exactly the three missing labels stated in the lemma; all other owners contribute all three cells. $\square$

At anchor (q, r) give labels 0, 1, 2 the respective integer energies

$$(3) \quad h_0(q, r) = q + r, \quad h_1(q, r) = -q, \quad h_2(q, r) = -r.$$

Lemma 3.2 (Local energy table). *Relative to the owner phase above, the four possible tile classes have the following energies.*

tile class	number of owners	energy
<i>in-phase right stone</i>	1	0
<i>wrong-phase right stone</i>	3	3
<i>two-owner bone</i>	2	1
<i>three-owner bone</i>	3	4

For a two-owner bone, the owner containing two cells is its source. The label missing at the source equals the label of the single target cell.

Proof. For an in-phase stone, the three energies at one owner sum to zero. The remaining cases depend only on the translation class modulo three. For a translate based at (i, j) , put

$$\rho \equiv i - j - t \pmod{3}, \quad \rho \in \{0, 1, 2\}.$$

Direct substitution in (3) gives

ρ	R	H	V	D
0	0	1	1	1
1	3	4	1	4
2	3	1	4	1.

Here the entry for R at $\rho = 0$ is the in-phase stone. The six energy-one bone cases have the following source-missing label and directed owner displacement:

bone and phase	label	step
$(H, 0)$	2	A
$(H, 2)$	2	B
$(V, 0)$	1	C
$(V, 1)$	1	B
$(D, 0)$	0	C
$(D, 2)$	0	A .

In every row, the two source cells are precisely the two labels other than the displayed one, and the target cell has the displayed label. Hence its energy is $-h_\ell(\text{source}) + h_\ell(\text{target}) = 1$. The three unlisted bone phases meet three owners and have energy four, which completes the table. $\square$

Proposition 3.3 (Energy rigidity). *Every right stone in a type-103 tiling of $\mathcal{B}(n, 2n - 3)$ is in phase, and every bone is a two-owner, label-preserving directed edge.*

Proof. Every full owner has total energy zero. At each corner the two present labels have total energy t , so Lemma 3.1 gives total benzel energy $3t = 3(n - 2)$.

Let s be the number of wrong-phase stones and f the number of three-owner bones. Lemma 2.1 gives $3(n - 2)$ bones. Of these, $3(n - 2) - f$ have energy one and f have energy four, while the s wrong-phase stones have energy three. The total tile energy is therefore

$$[3(n - 2) - f] + 4f + 3s = 3(n - 2) + 3(s + f).$$

Comparison with the benzel energy gives $s + f = 0$, hence $s = f = 0$. $\square$

4. THE DIRECTED Y

In anchor coordinates, the only directed displacements are

$$A = (2, -1), \quad B = (-1, -1), \quad C = (-1, 2).$$

In simplex coordinates they are

$$\begin{aligned} A &: (u, v, w) \mapsto (u - 1, v, w + 1), \\ B &: (u, v, w) \mapsto (u, v + 1, w - 1), \\ C &: (u, v, w) \mapsto (u + 1, v - 1, w). \end{aligned}$$

Label 0 permits A, C , label 1 permits B, C , and label 2 permits A, B . The corresponding potential h_ℓ increases by one on every edge carrying label ℓ . We replace each bone by the directed edge from its two-cell owner to its one-cell owner; the second table in the proof of Lemma 3.2 shows that this correspondence is unique in both directions.

Proposition 4.1 (Directed- Y bijection). *Type-103 tilings of $\mathcal{B}(n, 2n - 3)$ are in bijection with triples of labelled paths in Δ_t satisfying the following conditions.*

- (1) *The paths of labels 0, 1, 2 start at $(0, t, 0)$, $(0, 0, t)$, and $(t, 0, 0)$, respectively.*
- (2) *Each path uses its two permitted steps, and the vertex sets of two different paths intersect only at their common endpoint.*
- (3) *The paths have one common endpoint (u, v, w) with $u, v, w > 0$.*

Every full owner outside the three paths is occupied by a right stone.

Proof. An outgoing edge uses two cells of its source owner, whereas an incoming edge uses one cell of its target owner. At a full owner not occupied by a stone, let o and i denote the numbers of outgoing and incoming edges. Exact coverage gives $2o + i = 3$, and two outgoing edges would overlap. Hence either $(o, i) = (1, 1)$ or $(o, i) = (0, 3)$. In the first case the incoming edge covers the label omitted by the outgoing edge, so both carry the same label.

In the second case the incoming labels are 0, 1, 2. At a deficient corner the same argument gives $2o + i = 2$: either there is one outgoing edge carrying the missing label, or there are two incoming edges carrying the two present labels.

Since $|\Delta_t| = \binom{t+2}{2} = \binom{n}{2}$, Lemma 2.1 leaves

$$\binom{n}{2} - \frac{(n-5)(n-2)}{2} = 3t + 1$$

owners not occupied by stones. There are $3t$ bone edges. If K is the number of full three-incoming sinks and P the number of two-incoming corner sinks, every other active owner has one outgoing edge. Counting outgoing edges therefore gives

$$3t = (3t + 1) - K - P,$$

so $K + P = 1$.

No directed cycle is possible because the constant label potential strictly increases along every edge of a pass-through component. It remains to rule out $P = 1$. By cyclic symmetry suppose $(0, t, 0)$ is the corner sink. A label-1 path from $(0, 0, t)$ to this corner uses only B -steps: B leaves u unchanged, C increases it, and both endpoints have $u = 0$. Its vertex set is therefore

$$\{(0, b, t - b) : 0 \leq b \leq t\}.$$

A label-2 path from $(t, 0, 0)$ to the same corner must use exactly t A -steps and t B -steps. Immediately after its final A -step it is at $(0, b, t - b)$ for some $b < t$; equality $b = t$ would make the preceding w -coordinate negative. This vertex belongs to the label-1 path and is not the common corner sink, contradicting the local coverage rules. Hence $K = 1$ and $P = 0$.

Following outgoing edges from the three corners now gives the required three paths. They cannot meet before the unique sink: a pass-through owner has only one incoming edge, and the sink already has its three incoming labels. Their common endpoint is a full owner and therefore has three positive simplex coordinates. Any further component would have no sink and hence, by finiteness, would contain a directed cycle.

Conversely, replace every permitted labelled step by the unique bone from Lemma 3.2. At a source, the outgoing bone covers the two present labels. At a pass-through owner, the incoming bone covers the path label and the outgoing bone covers the other two labels. At the common sink, the three incoming bones cover labels 0, 1, 2. In-phase stones cover every unused full owner. The intersection condition ensures that these local coverings are compatible, so they form a unique tiling. $\square$

Write the sink as

$$(u, v, w) = (x + 1, y + 1, z + 1), \quad x + y + z = m = n - 5.$$

At the sink the last directions must be distinct, since equal directions would give the same predecessor owner to two paths. Respecting the two allowed

directions for each label leaves exactly two possibilities:

$$(+): (C, B, A), \quad (-): (A, C, B)$$

for labels $(0, 1, 2)$.

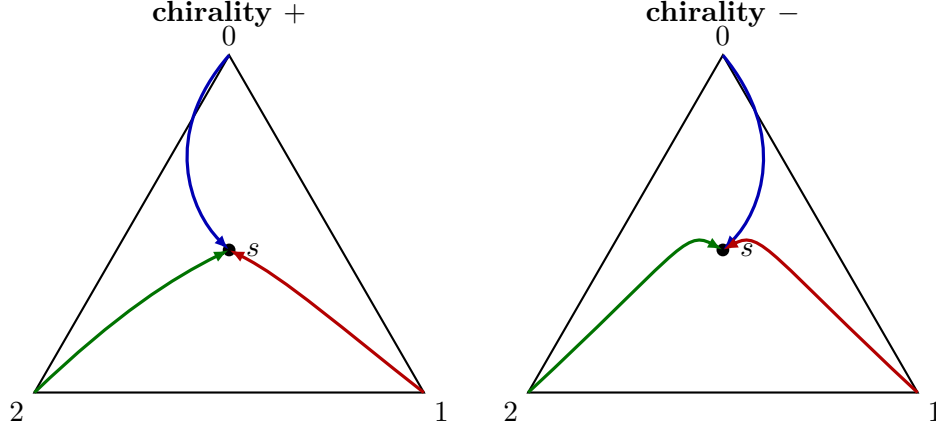


FIGURE 1. The three labelled arms, the fixed sink, and the two possible last-edge chiralities. The drawing is schematic; the proof uses the simplex coordinates.

5. FIXED-SINK ENUMERATION

Put

$$N_x = 2x + y + 2, \quad N_y = 2y + z + 2, \quad N_z = 2z + x + 2,$$

and adopt the convention $\binom{N}{-1} = 0$.

Lemma 5.1 (Independent arms). *Fix the sink $(x+1, y+1, z+1)$ and one of the two last-edge chiralities. Any three individually admissible labelled arms with those endpoints and last edges intersect only at the sink. The numbers of choices for the arms of labels $(0, 1, 2)$ are, respectively,*

$$\begin{aligned} (+): & \quad \binom{N_z}{z+1} - \binom{N_z}{z}, \quad \binom{N_x}{x+1} - \binom{N_x}{x}, \quad \binom{N_y}{y+1} - \binom{N_y}{y}, \\ (-): & \quad \binom{N_z}{z} - \binom{N_z}{z-1}, \quad \binom{N_x}{x} - \binom{N_x}{x-1}, \quad \binom{N_y}{y} - \binom{N_y}{y-1}. \end{aligned}$$

Proof. The step multiplicities from the three sources to the sink are

label	step multiplicities	
0	$A : z + 1$	$C : x + z + 2$
1	$B : x + y + 2$	$C : x + 1$
2	$A : y + z + 2$	$B : y + 1$.

For label 0, the only nontrivial simplex constraint along the path is $u \geq 0$, or equivalently that every prefix contains at least as many C -steps as A -steps.

For labels 1 and 2, the corresponding conditions are, respectively, at least as many B -steps as C -steps and at least as many A -steps as B -steps. Remove the prescribed last step. The reflection principle for a word with k minority letters gives $\binom{N}{k} - \binom{N}{k-1}$: reflecting the initial segment ending at the first negative prefix bijects the bad words with the words having $k - 1$ minority letters. This yields the six displayed counts.

It remains to check that independently chosen arms cannot meet prematurely. Every vertex on the label-0, label-1, and label-2 arms satisfies, respectively,

$$v \geq y + 1, \quad w \leq z + 1; \quad w \geq z + 1, \quad u \leq x + 1; \quad u \geq x + 1, \quad v \leq y + 1.$$

Thus a common vertex of arms 0 and 1 must lie on $w = z + 1$. In positive chirality the final B -step of arm 1 is its first arrival at this boundary; in negative chirality the final A -step of arm 0 is its first arrival. Hence the two arms meet only at the sink. The same argument on the boundaries $u = x + 1$ and $v = y + 1$, cyclically, separates the other two pairs. This proves both independence and the stated arm counts. $\square$

Corollary 5.2. *Let $P(x, y, z)$ and $Q(x, y, z)$ denote the positive- and negative-chirality counts at the fixed sink. Then*

$$(4) \quad P(x, y, z) = \prod_{\text{cyc}} \binom{2x + y + 2}{x},$$

$$(5) \quad Q(x, y, z) = \prod_{\text{cyc}} \left(\binom{2x + y + 2}{x} - \binom{2x + y + 2}{x - 1} \right).$$

In particular,

$$(6) \quad \frac{Q(x, y, z)}{P(x, y, z)} = \frac{(x + 3)(y + 3)(z + 3)}{(x + y + 3)(y + z + 3)(z + x + 3)}.$$

Proof. The negative-chirality product in Lemma 5.1 is exactly (5). For $N = 2r + s + 2$,

$$\binom{N}{r + 1} - \binom{N}{r} = \frac{s + 1}{r + 1} \binom{N}{r}.$$

Applying this identity to the three positive-chirality factors gives the additional multiplier

$$\frac{x + 1}{z + 1} \frac{y + 1}{x + 1} \frac{z + 1}{y + 1} = 1,$$

which proves (4). Finally,

$$\frac{\binom{2r+s+2}{r} - \binom{2r+s+2}{r-1}}{\binom{2r+s+2}{r}} = \frac{s + 3}{r + s + 3}.$$

Multiplying its three cyclic instances gives (6). $\square$

6. SUMMING THE SINKS

Define

$$P(t) = \sum_{x,y,z \geq 0} P(x,y,z)t^{x+y+z}, \quad Q(t) = \sum_{x,y,z \geq 0} Q(x,y,z)t^{x+y+z}.$$

Proposition 6.1. *Let $T = T(t)$ be the unique formal power series satisfying*

$$T = 1 + tT^3.$$

Then

$$(7) \quad P(t) = \frac{T^9}{(3-2T)(T^2-3T+3)},$$

$$(8) \quad Q(t) = (2-T)^3 P(t),$$

$$(9) \quad P(t) + Q(t) = T^9 \frac{3-T}{3-2T}.$$

Proof. Use

$$\binom{2x+y+2}{x} = [a^x](1+a)^{2x+y+2}$$

and its cyclic analogues in (4). Summing three geometric series gives (10): the powers of x , for example, form the geometric ratio $t(1+a)^2(1+c)/a$. Thus

$$(10) \quad P(t) = \text{CT}_{a,b,c} \frac{(1+a)^2(1+b)^2(1+c)^2}{D_a D_b D_c},$$

where

$$\begin{aligned} D_a &= 1 - \frac{t(1+a)^2(1+c)}{a}, \\ D_b &= 1 - \frac{t(1+b)^2(1+a)}{b}, \\ D_c &= 1 - \frac{t(1+c)^2(1+b)}{c}. \end{aligned}$$

We evaluate this formal constant term by the multivariate Lagrange–Good residue formula [4]. Its small-root equations are

$$a = t(1+a)^2(1+c), \quad b = t(1+b)^2(1+a), \quad c = t(1+c)^2(1+b).$$

They have a unique solution in $t\mathbb{Q}[[t]]^3$: once all coefficients below degree d are known, the equations determine the degree- d coefficients. Cyclic symmetry therefore gives $a = b = c = U$, where $U = t(1+U)^3$; put $T = 1 + U$.

At the small root, put

$$r = \frac{U}{T} = \frac{T-1}{T}, \quad d = 1 - 2r = \frac{2-T}{T}.$$

For $\phi_a = (1+a)^2(1+c)$ and its cyclic analogues, the Good denominator is the determinant of

$$\left(\delta_{ij} - \frac{x_j}{\phi_i} \frac{\partial \phi_i}{\partial x_j} \right)_{i,j \in \{a,b,c\}}.$$

At the small root this matrix and its determinant are

$$J = \begin{pmatrix} d & 0 & -r \\ -r & d & 0 \\ 0 & -r & d \end{pmatrix}, \quad \det J = d^3 - r^3 = \frac{(3-2T)(T^2-3T+3)}{T^3}.$$

The numerator in (10) becomes T^6 . Division by the determinant contributes the additional factor T^3 , proving (7).

The ballot identity

$$\binom{N}{k} - \binom{N}{k-1} = [a^k](1-a)(1+a)^N$$

with $\binom{N}{-1} = 0$ shows that the constant term for $Q(t)$ has the additional numerator $(1-a)(1-b)(1-c)$. At the small root this is $(1-U)^3 = (2-T)^3$, proving (8). Finally,

$$1 + (2-T)^3 = (3-T)(T^2-3T+3),$$

which gives (9). $\square$

Proof of Theorem 1.1. Differentiating $T = 1 + tT^3$ and simplifying (9) gives

$$P(t) + Q(t) = 2T^9 + \frac{1}{3}t \frac{d}{dt} T^9.$$

Hence

$$[t^m](P(t) + Q(t)) = \frac{m+6}{3}[t^m]T^9.$$

The ordinary Lagrange inversion formula yields

$$[t^m]T^9 = \frac{9}{2m+9} \binom{3m+8}{m}.$$

For $m = 0$ this is immediate; for $m > 0$ it follows from

$$[t^m](1+U)^9 = \frac{9}{m}[u^{m-1}](1+u)^{3m+8}.$$

Consequently

$$[t^m](P(t) + Q(t)) = \frac{3(m+6)}{2m+9} \binom{3m+8}{m}.$$

The adjacent-binomial identity

$$\binom{3m+8}{m-1} = \frac{m}{2m+9} \binom{3m+8}{m}$$

gives (1). Substituting $m = n - 5$ proves Theorem 1.1. $\square$

7. COMPUTATIONAL CHECKS

A direct exact-cover program implements the barycentric benzel and the four permitted prototile classes directly. It reproduces the formula for $5 \leq n \leq 11$ and records the owner graph of every tiling. A separate directed-path program verifies the fixed-sink chirality products and the total formula through $n = 12$. These computations are finite consistency checks and are not used in the proof of Theorem 1.1.

A kernel-only Lean 4 formalization defines the literal benzel and prototiles, proves the tile-count invariant and tiling/path equivalence, and verifies the generating-function and closed-form steps. Its exact 19-label map compiles with `-trust=0`; the endpoints use only `propext`, `Classical.choice`, and `Quot.sound`. The finite computations above remain independent consistency checks rather than formal premises.

Data and code availability. The complete formalization and reproducibility files are available in the [public code repository](#); the versioned software archive has DOI [10.5281/zenodo.22079396](#).

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE MANUSCRIPT PREPARATION PROCESS

During the preparation of this work, the author used an author-developed local implementation of the Proof Engine framework for AI-assisted mathematical research. The implementation operates through replaceable AI-agent and harness components; the present work used OpenAI Codex within that implementation. The framework is described in *Proof Engine Infrastructure* [5]. The author reviewed and edited the resulting material and assumes full responsibility for the definitions, statements, proofs, citations, code, and exposition.

STATEMENTS AND DECLARATIONS

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